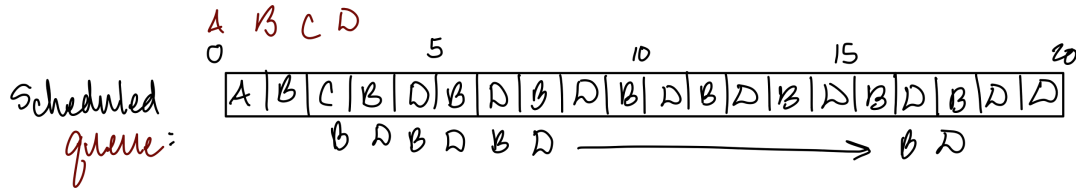
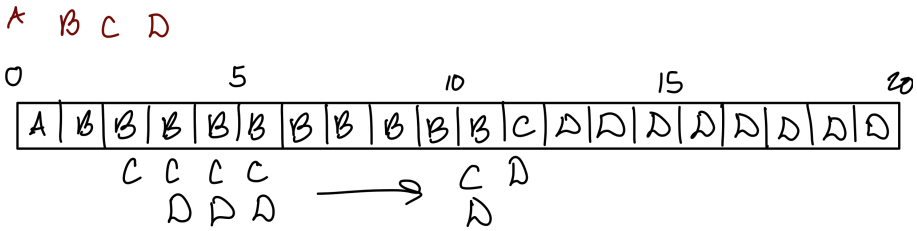
Destiny DeLara
Operating Systems
Assignment 3

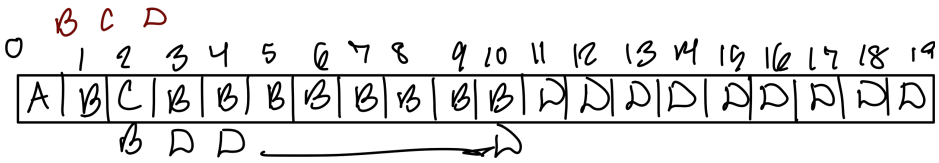
1. CPU Scheduling
   a. **RR, time quantum = 1**
   Round Robin is a preemptive scheduling algorithm, where each process is given a time quantum or time slice by the CPU. Each arriving process will be added to the tail of the ready queue. Processes will spend time in the CPU with this given time quantum, then it will either be released if it has finished its service time, or it will go back to the tail of the ready queue.



   b. **SPN**
   The Shortest Process Next algorithm is nonpreemptive. Here, the process with the shortest service time will get priority and get CPU time. For instance, if there is already a process in the CPU, and there are processes in the queue, the processes in the queue will have to sort based on their service time. Until the running process in the CPU is finished, then the next process that has the shortest service time will have CPU time next.

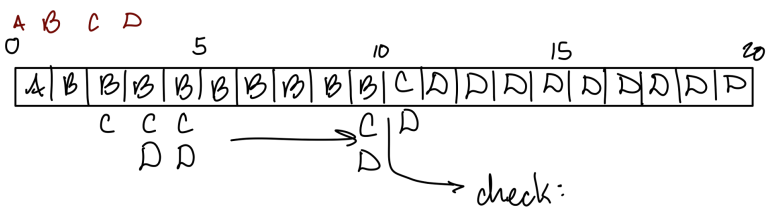


   c. **SRT**
   Shortest Remaining Time is the preemptive scheduling version of SPN. Every time a process arrives, the service time is compared. Rather than having the process wait until another is done with its service time, the arriving process with the shortest service time gets priority and swaps with the current process in the CPU. This process is moved to the queue if it isn't finished.



d. **HRRN**

The Highest Response Ratio Next algorithm is nonpreemptive and has priority for the process with the highest resulting value calculated with the equation, $R = \frac{w+s}{s}$. In this equation, w represents the process' waiting time which is the difference between the current CPU running time and the time the process has arrived. The service time is represented by s. Whenever the CPU is open, R is checked and calculated for every process waiting in the queue. Whichever process has the high value of R is decided to get CPU time and must finish before any other processes in the queue are to have their R values calculated again and compared.



check:

$w = \text{current} - \text{arrived}$

$R = \frac{w+s}{s}$

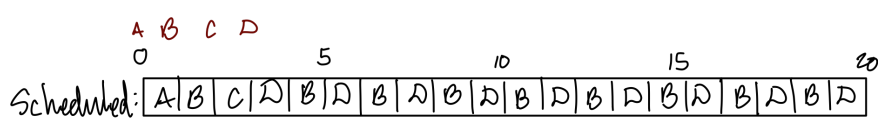
$R_c = \frac{10-2+1}{1} = 9$

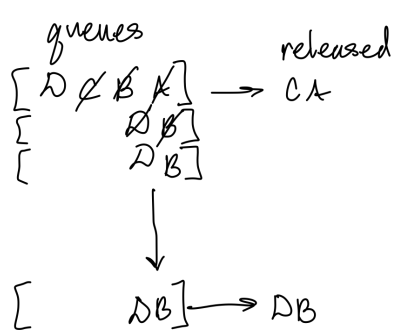
$R_D = \frac{10-3+9}{9} = 2$

$R_c > R_D$

e. **FB, time quantum = 1**

Feedback algorithm is preemptive with a given time quantum every process is allowed in the CPU. It uses a multi level queue scheduling method where processes move between priority levels based on CPU usage. Each queue is Round Robin Scheduled, except the lowest priority queue is scheduled with First Come First Serve. All newly arriving processes will start in the highest priority queue. Once a process finishes the time quantum, it will either move be released if it is finished with its service time or moved down to a lower priority queue. The next process to go into the CPU will be the process at the head of the highest nonempty priority queue.





2. Program output screenshots

Input:

```
CPU Scheduling Algorithms:
  a. First Come First Served (FCFS)
  b. Round Robin (RR)
  c. Shortest Process Next (SPN)
  d. Shortest Remaining Time (SRT)
  e. Highest Response Ratio Next (HRRN)
  f. Feedback (FB)
Select a scheduling algorithm (a-f): a
Enter a process name (or enter '0' to stop): A
Enter arrival time: 0
Enter processing time: 1
Enter a process name (or enter '0' to stop): B
Enter arrival time: 1
Enter processing time: 9
Enter a process name (or enter '0' to stop): C
Enter arrival time: 2
Enter processing time: 1
Enter a process name (or enter '0' to stop): D
Enter arrival time: 3
Enter processing time: 9
Enter a process name (or enter '0' to stop): 0
```

FCFS output:

```
Enter a process name (or enter '0' to stop): 0
You selected First Come First Served (FCFS)
Process execution order:
A
B
B
B
B
B
B
B
B
B
C
D
D
D
D
D
D
D
D
D
PS C:\Users\Desti\OneDrive\Desktop\cpu_schedule_
```

RR output:

```
You selected Round Robin (RR)
Enter time quantum: 1
Process execution order:
A
B
C
B
D
B
D
B
D
B
D
B
D
B
D
B
D
B
D
B
D
D
PS C:\Users\Desti\OneDrive\Desk
```

SPN output:

```
You selected Shortest Process Next (SPN)
Process execution order:
A
B
B
B
B
B
B
B
B
B
C
D
D
D
D
D
D
D
D
D
PS C:\Users\Desti\OneDrive\Desktop\cpu_sche
```

HRRN:

```
You selected Highest Response Ratio Next (HRRN)
Process execution order:
A
B
B
B
B
B
B
B
B
B
C
D
D
D
D
D
D
D
D
D
PS C:\Users\Desti\OneDrive\Desktop\cpu_schedule_algo
```

SRT:

```
You selected Shortest Remaining Time (SRT)
Process execution order:
A
B
C
B
B
B
B
B
B
B
B
D
D
D
D
D
D
D
D
D
PS C:\Users\Desti\OneDrive\Desktop\cpu_sched
```

FB:

```
You selected Feedback (FB)
Enter time quantum: 1
Process execution order:
A
B
C
D
B
D
B
D
B
D
B
D
B
D
B
D
B
D
B
D
```